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END TERM EXAMINATION
SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-112

Subject: Applied Mathematics-II

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Attempt all of the following questions:	(2.5×6=15)
(a)	Find all the points at which the mapping $w = (z + 1/2) / (4z^2 + 2)$ is not conformal.	(2.5)
(b)	Split i^i into its real and imaginary parts.	(2.5)
(c)	Find the Laplace transform of $t^2 e^{-2t}$.	(2.5)
(d)	Using the half range sine series of the function $f(x) = 1$ for $0 < x < \pi$, prove that: $1 + 1/3^2 + 1/5^2 + 1/7^2 + \dots = \pi^2 / 8$.	(2.5)
(e)	Taylor series expansion of $1 / (z - 2)$ in $ z < 1$ is	(2.5)
(f)	Classify the type of PDE: $\partial^2 u / \partial x^2 + 2(\partial^2 u / \partial x \partial y) + 5(\partial^2 u / \partial y^2) = 0$, whether it is parabolic, elliptic or hyperbolic.	(2.5)
UNIT - I		
Q2 (a)	Find all the values of $(-1 + \sqrt{3} i)^{3/2}$.	(7)
(b)	Verify that the function $u(x, y) = x^3 y - xy^3$ is harmonic and find the harmonic conjugate of $u(x, y)$ to express the function $f(z) = u + iv$ as an analytic function.	(8)
OR		
Q3 (a)	Evaluate the integral $\oint_C [e^{2z} / ((z - 1)(z - 2))] dz$ along the curve C, where C is the circle $ z = 3$.	(7)
(b)	Integrate the function $f(z) = \bar{z}$ along the curve C, where C is the square with vertices $z = 0, 2, 2i, 2 + 2i$.	(8)

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Q. No.	Question	Marks
UNIT - II		
Q4 (a)	Find the bilinear transformation (Möbius transformation) which maps 1, i, -1 of the z-plane onto 1, i, -1 of the w-plane respectively. Also find the fixed points or invariant points.	(8)
(b)	Find the singular points, type of singularities and the corresponding residues of the function $f(z) = 1 / (z^2 - 1)^2$.	(7)
OR		
Q5 (a)	Sketch and graph the given region: $z \leq 1/2$, $-\pi/8 < \text{Arg}(z) < \pi/8$, and its image under the mapping $w = z^2$.	(7.5)
(b)	Prove that the integral $\int_{-\infty}^{\infty} [x^2 / ((x^2 + 1)(x^2 + 4))] dx = \pi / 3$.	(7.5)
UNIT - III		
Q6 (a)	Using Laplace transform solve the ordinary differential equation: $y'''(t) + 2y''(t) - y'(t) - 2y = 0$, with conditions $y(0) = y'(0) = 0$, and $y''(0) = 6$.	(8)
(b)	Find the inverse Laplace transform of $\log[(s + 1) / (s - 1)]$.	(7)
OR		
Q7 (a)	Find the Fourier series to represent the function $f(x) = x^2$ in the interval $(-\pi, \pi)$.	(7.5)
(b)	Find the Fourier transform of $f(x) = 1$ for $x < 1$, and 0 for $x > 1$. Hence evaluate $\int_0^{\infty} [(\sin x) / x] dx$.	(7.5)
UNIT - IV		
Q8 (a)	Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature zero, assuming that the initial temperature is: $f(x) = x$ for $0 < x < L/2$, and $L - x$ for $L/2 < x < L$.	(8)
(b)	A tightly stretched string with fixed end points $x = 0$ and $x = l$, is initially in a position given by $y = \sin^3(\pi x / l)$. If it is released from rest from this position, find the displacement $y(x, t)$.	(7)
OR		
Q9 (a)	Solve the Laplace equation $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$, subject to the conditions: $u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, a) = \sin(n\pi x / l)$.	(8)
(b)	Solve the equation $\partial u / \partial t = \partial^2 u / \partial x^2$, with boundary conditions: $u(x, 0) = 3 \sin(n\pi x)$, $u(0, t) = 0$, and $u(1, t) = 0$, where $0 < x < 1$, $t > 0$.	(7)

**** End of Question Paper ****